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Review

Progress in the Application of Laser-Induced Breakdown Spectroscopy in Coal Quality Analysis

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Abstract: In recent years, with the increase in environmental awareness, people have become more and more concerned about the effectiveness with which coal burns. Laser-induced breakdown spectroscopy (LIBS) has become an important way of coal elemental analysis because of its uncomplicated sample handling, remote sensing capability, and superior sensitivity in identifying a wide range of elements, including both major and minor constituents, down to trace levels. However, the complexity of its mechanism of action, the experimental environmental factors, and the presence of matrix effects in its measurement spectrum have affected the measurement accuracy. In this paper, on the basis of introducing the experimental process and principle of LIBS, we summarize and analyze the influence of each factor on the LIBS detection medium, summarize the mainstream model analysis algorithms, and analyze the advantages and disadvantages of each model. While summarizing the LIBS in media detection in recent years, it aims to provide strong support and guidance for subsequent more in-depth exploration and research.

Keywords: LIBS; coal; experimental principles; analysis of experimental errors; data processing models

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1. Introduction

Energy is a key driver of sustainable economic development and an essential material for human survival [1]. And coal is one of the world's most important primary energy sources, accounting for 27% of global primary energy consumption according to BP World Energy Statistics 2021 [2]. In recent years, the world has only partially reduced its supply of fossil energy, but socioeconomic life has been greatly impacted, which also demonstrates the need for an orderly low-carbon energy transition [3]. Coal will remain irreplaceable for some time to come.

According to the characteristics of coal quality, its analysis methods are mainly elemental analysis (carbon, hydrogen, oxygen, nitrogen, sulfur, etc.) and industrial analysis (ash, volatile matter, moisture, and fixed carbon and calorific value). As coal is affected by mining, transportation, storage, and climatic conditions, there are great differences in the content of each component of coal. For coal-fired burners, one of the pressing issues they confront is the imperative need to stabilize flame combustion, as failure to do so hinders proper ignition, efficiency, quenching processes, and pollutant management [4]. For coal-fired power plants, in order to improve the combustion efficiency of the boiler, reduce the generation of unburned carbon, reduce the emission of pollutants, and provide the necessary reference for boiler operation and design, the analysis of ash, volatile matter, and calorific value are of great significance for the optimal utilization of the coal mixture. In terms of industrial analysis of coal, coal ash is the mineral residue left after complete combustion; the higher the ash content, the lower the effective carbon content of coal, the lower the calorific value, and the increase in non-combustible content [5]. Volatiles

are thermally decomposed products stemming from the organic matter present in coal, primarily comprising hydrocarbons, hydrogen gas, and carbon monoxide. A high degree of volatility, in conjunction with a high caloric value, can exacerbate slagging and lead to the deterioration of nozzles. On the other hand, a low volatile content, when paired with a high ash content, results in coal that is inefficient in terms of its combustion properties [6]. Energy content serves as a pivotal metric in assessing coal consumption, thermal efficiency, heat equilibrium within combustion equipment, as well as varying coal proportions [7]. It is, therefore, crucial to measure the composition of coal quickly and accurately.

The traditional analysis of coal quality indexes, usually offline coal ash analysis, takes a considerable amount of time from sampling and sample preparation to testing. For a long time, the traditional high-temperature furnace insulation air heating method has been used to measure the volatile fraction of coal combustion; this offline method, with a large workload, small data collection, and long analysis time, generally measured the volatile fraction of coal combustion than the actual boiler operating conditions lagged behind for more than a few hours [8]. There is also the chemical analysis method, in which the content of various elements in coal is determined by reacting chemical reagents with specific components in the coal sample. For example, the weight method can be used to determine the ash and moisture in coal, the volumetric method can be used to determine the sulfur, and the colorimetric method can be used to determine certain oxides in the ash as well as the sulfur. This method has high accuracy and reliability but requires complex operation processes and specialized chemical knowledge. Although these two methods have high accuracy, they have shortcomings such as long analysis cycles and poor real-time performance, which make it difficult to meet the demand for rapid analysis in industrial applications and cannot meet the requirements for optimal control of equipment [9].

To achieve swift or immediate monitoring of ash content, volatile components, and thermal energy potential, as well as to refine coal blending strategies and enhance the combustion efficiency of coal-fired power stations, the prompt gamma-ray neutron activation analysis (PGNAA) technology stands as the premier online coal quality assessment tool within the coal industry. The principle of PGNAA is to use neutron bombardment of coal particles to generate gamma rays for analysis, which can be generated using reactors, gas pedals, or isotope neutron sources [10]. Nonetheless, variations in the characteristics of coal and modifications to the measurement settings may compromise the precision of PGNAA measurements, while also posing potentially expensive nuclear radiation risks [11]. Near-infrared spectroscopy (NIRS) is capable of recognizing the quality and composition of coal, albeit its utilization is constrained by the feeble nature of coal's diffuse reflectance signals and the demanding conditions prevalent in coal-fired power stations [12]. X-ray fluorescence spectrometry (XRF) enables immediate analysis of coal composition. Nevertheless, because of the constraints imposed by the instrument's capabilities and the scarcity of X-rays emitted by lighter elements, its detection range is limited to elements possessing atomic numbers greater than 11 [13]. Dual-energy gamma-ray transmission serves as a prevalent online tool for quantifying coal ash, but its practicality is hindered by factors such as substantial equipment size, the inherent risk posed by nuclear radiation, and elevated operational expenses [14]. While atomic absorption spectrometry (AAS) boasts remarkable sensitivity, its implementation necessitates intricate preprocessing steps [15], and it also lacks the ability to identify several elements at once. If knowledge of several elements in a sample is desired, an analysis must be run for each sample separately. This leads to an increase in the time required to complete the analysis [16]. Inductively coupled plasma-atomic emission spectrometry (ICP-AES) offers high-accuracy coal analysis capabilities, yet necessitates a lengthy and intricate process for sample dissolution [17]. The sample preparation often requires multiple dilution steps, and the cost of the instrument is relatively high and not always available. In summary, the various techniques described above have a variety of shortcomings that prevent them from realizing fast, safe, and accurate coal measurements [18].

As laser technology and spectroscopic detection methods have advanced, laser-induced breakdown spectroscopy (LIBS) has emerged as a novel analytical tool. This technique employs a pulsed laser beam, typically ranging from tens to hundreds of millijoules in energy, to ignite a plasma on the surface of a sample. Subsequently, spectroscopic analysis instruments are utilized to decipher atomic, ionic, and molecular spectral signatures from the plasma, enabling both qualitative identification and quantitative measurements of the sample's composition. Not only does it enable an almost unlimited range of elemental detection, the sample preparation process is simple and fast, with few complex steps required. Compared to other technologies, LIBS' non-contact detection ensures human safety, while its simple sample preparation and fast and direct analytical capabilities allow for the efficient and non-destructive detection of multiple elements simultaneously. In addition, the technology has the potential to be used remotely and can be applied to a wide range of samples, including solids, liquids, and gases, demonstrating its flexibility and adaptability. Overall, the advantages of LIBS include simple sample preparation, telemetry capabilities, high sensitivity for multi-element detection, and the ability to analyze heavy and light elements and even some trace elements [19]. There have been many studies and applications of LIBS in alloys, rocks, fossil fuels, environment, and food safety [20]. However, the raw spectral data do not directly yield the desired experimental results, and mining the raw spectral data to obtain as much valid information as possible becomes the next step.

There are many types of coal in the world, and their elemental contents and types show significant differences depending on their types and origins, which directly affect the combustion performance, industrial applications, and environmental impacts of coal. From the elemental analysis of coal, it can be seen that the composition of coal mainly contains C, H, O, N, S, Al, Si, Fe, Ca, and more than 60 kinds of trace elements, which is more complicated than most of the compounds or mixtures in terms of the number of elements it contains. For LIBS to analyze coal composition, how to convert spectral information into elemental composition is an important task.

A considerable number of researchers have delved into the potential applications of LIBS technology for identifying coal elements and assessing coal quality. In the landmark year of 1990, Ottesen and their team ventured into using LIBS technology for analyzing the composition of coal dust, conducting a qualitative assessment of various elemental constituents; this pioneering effort marked the initial exploration into the viability of employing LIBS technology as a means to quantify coal dust, paving the way for future advancements in this field [21]. Body et al. developed a test device for rapid analysis of LIBS, optimized the data analysis method, and achieved simultaneous analysis of major inorganic elements and some organic elements in coal, further proving the accuracy and reproducibility of the analysis [22]. Gaft et al. designed a LIBS detection device for online monitoring of ash content in coal, with measurements that meet the user's needs [23]. Lu et al. investigated the total calorific value, carbon content, volatile content, and ash content of coal using the partial least squares (PLS) method [24]. Wang et al. proposed a quantitative analysis algorithm that combines dominant factors with partial least squares to compute coal components, with results that outperform traditional partial least squares models [25]. Many chemometrics methods have been developed for sample classification and identification in conjunction with LIBS, ranging from early principal component analysis (PCA) and partial least squares (PLS) algorithms to machine learning due to the rise of artificial intelligence, deep learning, and migration learning based on small samples, etc., all of which have been used with good results in LIBS applications.

At the present time, a tutorial review on coal analysis has been published, summarizing recent advances in the decade 2011–2020 [26]. However, with the development of artificial intelligence in the last two years, more deep learning neural networks have been developed, many of which have achieved good results in LIBS analysis of coal. In this paper, we focus on systematically introducing the principles and processes of the LIBS system, analyzing the possible sources of errors at the experimental and principle levels, and the

advances and comparisons of each major analytical model. On the basis of analyzing and summarizing the current research on LIBS, we aim to discuss in depth the latest progress and application prospects of the technology so as to provide a series of feasible insights for subsequent researchers on how to conduct in-depth studies using LIBS technology, including experimental design, data analysis, and other aspects.

2. LIBS Experimental Procedure and Principle

2.1. Experimental Procedure

The LIBS hardware system consists of a Q-switched Nd: YAG laser, a spectrometer, a laser attenuator, a reflector, a plano-convex lens, a laser energy meter, a mobile platform, an optical flat panel, and a control computer, as shown in Figure 1, which represents only the main instrumentation of the test system. In particular, a Q-switched Nd: YAG laser was used as a light source to generate a parallel laser beam with a wavelength of 1064 nm, an energy of 300 mJ, and a frequency of 5 Hz. This laser beam is then precisely tuned by a laser attenuator, which attenuates the energy to a stabilized 90 mJ (measured energy fluctuation is less than 1%). Then, the parallel laser was changed to a vertical direction through a 45-degree reflector, focused by a UV fused silica plano-convex lens with a focal length of 75.3 mm, and acted directly on the surface of the coal cake to excite the coal plasma. The spectral signals emitted by these plasmas were efficiently collected by two plano-convex lenses with a focal length of 40.1 mm, subsequently focused at the fiber optic flange, and transmitted to the spectrometer for spectral analysis. The elemental composition can be determined by comparing the peaks of the presented spectral image with the wavelengths of atomic spectra from the National Institute of Standards and Technology (NIST). Ultimately, the LIBS spectral data are accurately recorded and stored by a computer for subsequent data processing and analysis. In the system, the laser energy meter is responsible for monitoring the fluctuation of the laser energy in real time to ensure the stability of the spectral data; the mobile platform is used for multi-point LIBS spectral acquisition on the surface of the coal cake to cope with possible inhomogeneities in the cake matrix; and the control computer is not only used to collect the spectral data but also to accurately set up the triggering delay time of the laser and the spectrometer to meet the different experimental needs. The efficient collaboration of the whole system provides strong support for the in-depth research and application of LIBS technology in coal and other fields.

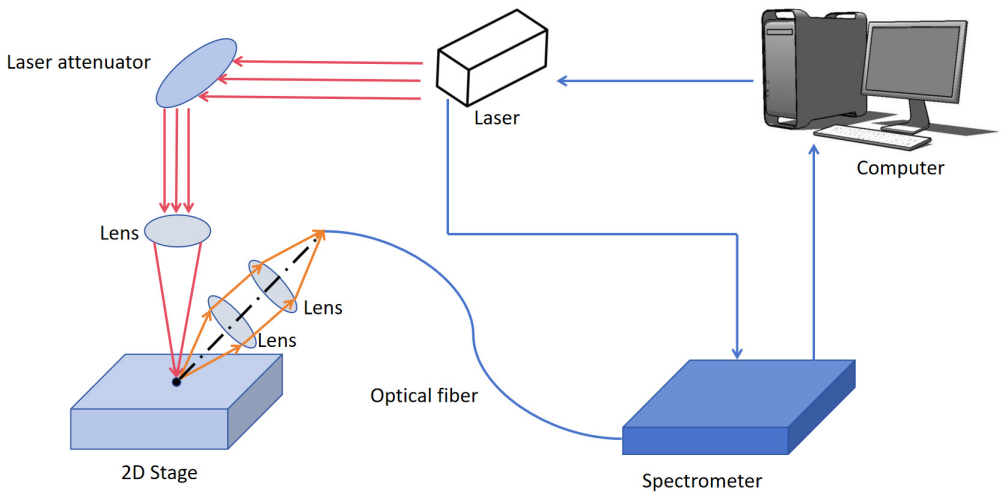

Figure 1. Schematic of the LIBS experimental setup in the laboratory.

In addition, lasers commonly used in LIBS technology are diode-pumped solid-state lasers (DPSSL) [26], transversely excited atmospheric (TEA) CO_2 lasers, fiber lasers (FL), and more. DPSSL uses a diode as the pump source, which has the advantages of high efficiency, stability and long lifetime, and is one of the commonly used lasers in LIBS

technology. Compared to the commonly used Nd: YAG laser, a TEA CO_2 laser emits radiation at a longer wavelength (mid-IR) and has a longer pulse duration [27]. These laser characteristics favor the analysis of powder samples [28]. However, due to limitations such as their long wavelengths and low energy conversion efficiencies, gas lasers have been used relatively infrequently in LIBS technology. FL is a new type of laser that has been developed rapidly in recent years, which not only has high particle number inversion efficiency and good beam quality but also has a series of advantages such as miniaturization, low cost, good stability, no water cooling, and only simple air cooling [29]. Currently, the Q-switched Nd: YAG laser is still the mainstay, and this paper will use this laser as the basis for introducing various aspects of LIBS.

2.2. Measurement Principle

LIBS is an atomic emission spectrometry analysis technique in which the surface of a coal sample is ablated by a laser to form a plasma with strong brightness, which is received by an optical fiber and transmitted to a spectrometer to complete the collection of plasma spectral signals, and then a quantitative model is built for the coal sample with both spectral data and coal quality data, and the prediction results can be obtained by bringing the LIBS spectra of the unknown coals into the built model.

(1) The process of plasma formation

The plasma formation process is through the pulsed laser through the lens and reflector focused on the sample surface in the high intensity of the laser pulse under the action of the sample surface particles will absorb the laser burning energy from the solid fusion state into a vapor state, and finally form a plasma. At the same time, the plasma expands rapidly due to the increase in temperature, and the surface of the coal sample decomposes into small irregular particles due to the thermal pressure [30]. After the end of the pulsed laser action, the plasma temperature decreases continuously, and during the plasma cooling process, the particles in the excited state jump down to the low energy level or the ground state, and their excitation energy is radiated outward in the form of light, which produces the characteristic spectral lines corresponding to the elements, and the unstable electrons of the high-energy state emit spectra of a certain wavelength when they jump back to the low energy level or the ground state. The electrons jump between discrete bound energy levels to form linear spectra, i.e., the characteristic spectral lines of atoms or ions, and the wavelengths and intensities of the spectral lines represent the species and concentration of the elements, respectively. Finally, the plasma spectra are collected by a spectrometer and transmitted to a computer, where the wavelengths and intensities of the characteristic spectral lines can be analyzed to obtain information on the types and contents of the elements [31].

As shown in Figure 2, there are three main regions of the plasma from the bottom up: the first region is the plasma heat core, the second region is the Knudsen layer between the sample surface and the heat core, and the third region is the outermost part of the plasma. Finally, due to the strong collision effect between the plasma and the air, a shock wave will be formed at the foremost part of the plasma plume [32].

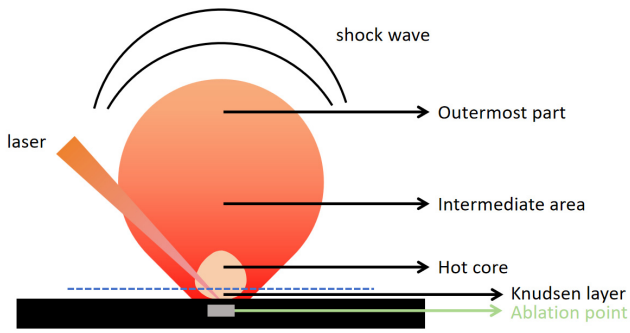

Figure 2. Schematic structure of the plasma.

(2) Determination of peak atomic radiation

The spectral picture formed by LIBS applied to medium detection contains many peaks, and the peaks are marked by finding the local maxima in the data series and determining whether the point is an inflection point. Subsequently, the center wavelength of the recorded peaks is compared with the wavelength of the atomic spectrum of the NIST. Aligning the peak wavelengths with the excitation probabilities and actual elemental content in the coal mitigates the issue of peak center wavelength drift resulting from spectrometer resolution constraints, facilitating the precise identification of the coal cake's elemental makeup based on the waveform peaks [32].

2.3. Spectral Preprocessing

(1) Spectral baseline removal

Due to the presence of environmental conditions and equipment interferences, the baseline of the averaged spectra is often shifted, and this shift can directly affect the accuracy of the spectra and the accuracy of subsequent analysis results. Therefore, baseline correction or baseline removal can effectively improve the spectral signal-to-noise ratio (SNR). The findings indicate that the adaptive iterative reweighting penalized least squares approach (airPLS) effectively mitigates background noise, achieving superior results in comparison [33]. Without requiring any manual input or prior knowledge, the process relies solely on the iterative adjustment of weights based on the squared error difference between the estimated baseline and the original signal, offering a swift and adaptable solution [34].

(2) Spectral normalization

Spectral normalization, as a pivotal data processing methodology, aims to diminish or eradicate fluctuations within spectral datasets, thereby facilitating the comparison and examination of experimental conditions across these refined data. The use of normalization, trend separation, and correction ensures that spectral data acquired under different experimental conditions can be compared on a uniform scale.

Specifically, the normalization process scales the spectral data to a standard in order to better show the features in the spectrum; the trend separation technique is used to eliminate any long-term or systematic trends or drifts that may exist in the spectrum; and the correction step addresses errors caused by instrumental, environmental, or other factors. The formulas are as follows:

$$x_{z-scores} = \frac{x - \mu}{\sigma}$$

where $x_{z-scores}$ is the normalized spectral intensity, x is the spectral intensity, μ is the average spectral intensity of all spectra at the current wavelength, and σ is the standard deviation of the spectral intensity of all spectra at the current wavelength [32].

Not only do the aforementioned two methods elevate the comparability of spectral data, but they also broaden their utilization in diverse scientific research and application domains. This enhancement allows experimentalists to interpret and contrast spectral data with increased accuracy and reliability.

2.4. Experimental Sample Preparation

The phenomenon of locally different elements in coal blocks is common and is related to a variety of factors, such as its formation process, the degree of coalification, the geological environment, and post-processing. For LIBS, its precision and accuracy depend largely on the homogeneity of the samples, and an overall elemental composition cannot be obtained by direct testing of coal briquettes. For pulverized coal samples, the structure is loose, there are a lot of gases inside, and the surface is not flat; if directly excited by the laser will be splashed and will act on the air inside the pulverized structure, which is not conducive to the collection of spectra.

The appeal analysis shows that when the medium is in powder, it is not favorable for the compositional detection of LIBS, so it can be carried out by making the pulverized coal into coal cake. Coal cake samples are usually made by pressing the pulverized coal with a press machine to form a coal cake with good mechanical properties for testing. On the one hand, the elemental content of the homogeneously mixed pulverized coal tends to be the same at all points inside the pulverized coal, and on the other hand, the combustion of pulverized coal has a higher combustion efficiency and less ash production. LIBS-based quantitative analysis of coal cake, for example, consists of several main processes: first, the spectra of each calibration sample are collected by multiple laser ablations; second, the spectra of each sample are averaged one by one. The dataset with a large sample size is used to build a prediction model, and the analysis of this stage is listed in Section 3 due to the fact that the differences in the composition of various coal cakes, experimental instrument parameters, experimental environment, and other factors can have a large impact on the experimental results. Subsequently, the model was applied to a real industrial environment. If certain predictions are frequently outliers, the model is usually modified and corrected. Logically, the accuracy and robustness of the predictions depend on the performance of the model, and better prediction models require even hundreds of samples for calibration. In the field of quantitative analysis of coal by LIBS, many researchers have proposed various highly accurate predictive models for quantitative analysis of the coal industry based on a large number of training samples [35]. Their measurement accuracy and their respective advantages and disadvantages are listed in Section 4.

3. Experimental Error Analysis and Spectral Preprocessing

Realizing the rapid detection of medium elements and obtaining various parameters of coal in time through the composition analysis can realize the real-time control of the boiler, effectively guide the combustion process, improve combustion efficiency, reduce boiler slagging, reduce the unburned carbon and flue gas emissions, to improve the thermal efficiency of the boiler to reduce environmental pollution has a significant help. How to quickly and accurately obtain the composition of the media has become a topic of concern. LIBS has the ability of rapid detection, but the premise of LIBS technology to accurately analyze coal is to obtain plasma spectra that can accurately reflect the information of its components. At the same time, LIBS experiments are more linked and influence parameters and will significantly affect the stability of the spectra, and the reproducibility of the experimental parameters is very necessary to optimize. In the following, we will analyze the influence of several key experimental system parameters on the plasma spectral signal from the processes of raw material preparation, plasma generation, and spectral acquisition so as to realize a more accurate measurement of the medium as much as possible.

3.1. Principle Section

(1) Enhanced LIBS Plasma Strength

(I) Dual pulse configuration

The intensity of LIBS spectral signals can be bolstered through the adoption of a double-pulse setup. This configuration involves the employment of two laser pulses, with inter-pulse delays spanning from nanoseconds to microseconds, to induce plasma formation. These pulses may originate from a solitary laser source or be derived from two distinct lasers. Four prevalent geometrical arrangements employed in the double-pulse configuration include collinear alignment, intersecting beams, orthogonal pre-ablation, and orthogonal reheating setups, each offering unique advantages for spectral enhancement [36]. In the collinear configuration, the two laser beams travel parallel to each other along a shared axis, maintaining an orthogonal orientation relative to the sample surface [37]. In the cross-beam configuration, the two laser beams intersect at an angle, converging precisely on the target surface [38]. In the orthogonal configuration, beams cross orthogonally, one parallel to the sample, the other orthogonal. When a parallel beam ignites air plasma over the target, and orthogonal ablates the sample, it is known as a pre-ablation orthogonal setup [39].

When a quadrature beam is concentrated on a target to generate plasma and a subsequent parallel beam is directed with a delay to revitalize the plasma, it is termed as a quadrature reheating configuration [40]. During the laser–material interaction, the double-pulsed laser acts on the sample surface by both preheating and pre-ablation [41]. In comparison to single-pulse LIBS, dual-pulse LIBS is capable of transferring greater laser energy into the ignited plasma [42]. Furthermore, double-pulse LIBS mitigates plasma shielding of the laser, enabling longer luminescence duration of the excited plasma [36]. Consequently, enhanced plasma plumes and intensified plasma emission signals are achievable [42].

Wang et al. discovered that the gaseous atmosphere created by the initial laser beam mitigates the self-absorption of spectral signals from Cu, Mn, and Ni elements excited by the subsequent laser beam. They further demonstrated that spectral self-absorption strongly correlates with the double-pulse delay, assuming the constant energy of the first beam while exhibiting a weaker link to variations in the second beam's energy [43]. Michela Corsi et al. found that analyzing contaminated soil with coaxial dual-pulse LIBS enhanced the LIBS signal in easily compacted samples by a factor of 5–10 compared to single-pulse LIBS [44]. V. Piñon et al., by employing co-linear geometry double-pulse femtosecond LIBS on brass, iron, silicon, barium sulfate, and aluminum samples, the LIBS signals were enhanced by a factor of 3 to 10 when optimal time intervals were applied between the two ablation pulses [45].

(II) Resonance-enhanced laser-induced breakdown spectroscopy

Resonance-enhanced laser-induced breakdown spectroscopy (RELIBS) is a spectroscopic technique used to analyze the composition of samples. It combines LIBS and resonance-enhanced techniques with the aim of improving the sensitivity and selectivity for the detection of specific elements in a sample [46].

In RELIBS, resonance enhancement is realized by tuning the frequency of the laser to the resonance frequency of the target element as compared to ordinary LIBS. When the laser frequency matches the resonance frequency of the target element, the interaction between the laser and the target element is significantly enhanced. This enhancement can lead to a significant increase in detection sensitivity. By analyzing the spectrum emitted by the plasma, the elements present in the sample and their concentrations can be determined. Under resonance enhancement conditions, the signal of the target element is significantly enhanced, thus increasing the sensitivity and selectivity of the detection. RELIBS technology has been used in a wide range of applications in many fields, including environmental monitoring, materials science, geology, chemical analysis, etc. The RELIBS technology has virtually no limitation on the sample morphology, allowing the analysis of solid, liquid, and even gaseous samples, and since only a brief heating of the sample is required, using the RELIBS is a non-destructive analytical technique as it only requires a short heating of the sample by laser pulses [46]. When small amounts of material are ablated, the advantages of RELIBS over LIBS become apparent, and less damage is caused to the sample [47].

(III) Nanoparticle-enhanced laser-induced breakdown spectroscopy

Nanoparticle-enhanced laser-induced breakdown spectroscopy (NELIBS) is an optical emission method that relies on the deposition of plasma-generated nanoparticles onto a sample's surface via the laser-induced plasma (LIP) process. Due to the large specific surface area and surface enhancement effect of nanoparticles, they can effectively capture the laser energy and convert it into thermal energy, and when interacting with the laser, they absorb the laser energy and rapidly heat up, generating a localized microenvironment of high temperature and high pressure [48]. In this technique, nanoparticles are introduced into the sample to be analyzed to enhance the effect of laser-induced breakdown, thereby increasing the intensity of the spectral signal and detection sensitivity. Compared to signals obtained using conventional LIBS, NELIBS has better applicability to a wide range of sample types and morphologies, including solid, liquid, and gaseous samples, and the use of plasma nanoparticles enhances the signal, resulting in exceptional sensitivity and extremely low detection thresholds when compared to LIBS performance [49].

(IV) Flat top laser beam profile

In the case of conventional lasers, the laser emission follows a Gaussian distribution. Research has demonstrated that employing flat-topped laser beam profiles, characterized by a uniform energy distribution, enhances the effectiveness and efficiency of laser interactions [50]. Furthermore, the flat-top profile boasts a more balanced intensity distribution, which channels greater energy towards the ablation process, thereby mitigating material heating and ultimately elevating ablation efficiency [51]. An additional advantage of utilizing a flat-topped laser beam profile lies in its increased resilience to pulse energy fluctuations. This translates to a more consistent ablation spot diameter compared to a Gaussian beam, leading to a more stable and predictable laser processing outcome [52].

(2) Self-Absorption of Spectral Lines

During the emission of light from the heated interior of a LIP transitioning to the cooler exterior, the radiant energy can be reabsorbed by similar atoms and molecules that initially emitted it. Consequently, the spectral data captured by the spectrometer undergoes attenuation, compounded by the plasma's inherent absorption of its own radiation—a phenomenon known as self-absorption. This specific type of absorbing source diminishes peak intensities and broadens the plasma profile. Intriguingly, in certain spectral line configurations, the central portion of the line undergoes greater absorption than its flanks, giving rise to a distinctive pattern of self-absorption known as self-reversal [53]. For atomic lines characterized by high energy levels and low excitation thresholds or spectral lines exhibiting high transition probabilities, self-absorption becomes pronounced. Fortunately, these spectra can undergo correction to mitigate self-absorption effects, enhancing the linearity of LIBS calibration curves. Notably, self-absorption coefficients are defined in various manners, encompassing the comparison of peak heights, line area ratios, and the proportion of signal widths, each offering insights into the intensity of self-absorption [54].

During experimental procedures, the non-uniformity of laser-induced plasma gives rise to a phenomenon where high-energy particles within the plasma emit spontaneous radiation that is subsequently absorbed by lower-energy particles along its path. This interaction results in a reduction in spectral intensity recorded during the experiment [55]. In scenarios where a pronounced temperature gradient exists between the interior and exterior of the laser-induced plasma (LIP), the central region of spectral lines may experience more severe absorption than their peripheral counterparts, potentially causing a dip to manifest in the spectrum at the center wavelength. This self-absorption effect introduces distortions into experimentally obtained spectral lines, thereby compromising the accuracy of quantitative analyses performed on these spectra [56].

In order to eliminate or minimize the error of its self-absorption effect on the measurement results, many researchers in various fields have carried out a lot of work by investigating the influence of ambient pressure, ambient gases, and the use of double-pulse configurations on the self-absorption effect.

Tang et al. demonstrated that microwave-assisted excitation laser-induced breakdown spectroscopy (MAE-LIBS) can effectively reduce the self-absorption of the Na and K spectral lines and a little bit for the spectral lines with weak self-absorption and that it can simultaneously reduce the self-absorption of multiple elements [57]. Hao et al. investigated the effect of air pressure on the self-absorption of Cu and Mn spectral lines, and the results demonstrated that the self-absorption was substantially mitigated at 1 kPa [58]. Under their different ambient pressures, Ke et al. found that the self-absorption effect of the spectral line Cu I 521.82 nm is enhanced with increasing delay time and laser energy in both high and low vacuum environments, which is explained by the fact that the increase in delay time and laser energy leads to an increase in the plasma column density [56].

Wang et al. investigated the effect of air gas types (air, Ar, and N_2) on Al alloys by comparing the intensity of spectral lines with and without the self-absorption effect on the self-absorption effect; the experimental results show that Ar significantly enhances the self-absorption of the spectral lines compared to air and N_2 [59]. Rezaei et al. numerically

calculated the spectra of He and Ar in the noble gas after plasma formation and investigated the effect of Ar and He on the self-absorption of different Al spectra, which showed that the self-absorption of Ar is more severe than that of He [60]. However, Zehra K et al. investigated the effect of pressure on the intensity of Si spectral lines in Ar, He, and Ne, respectively, and found that the highest intensity was found in Ar, followed by Ne, and the lowest intensity was found in He [61]. Bian et al. found that a high signal-to-noise ratio could not be obtained for the C I 193.09 nm spectrum in an air atmosphere. Under argon atmosphere, the background of the C I 193.09 nm line was smoother, and the intensity of the line was significantly improved. At the same time, the signal-to-noise ratio of the spectrum obtained under argon atmosphere was also significantly improved, indicating that argon atmosphere can significantly improve the signal-to-noise ratio of the spectrum [62]. Through the above literature analysis, Ar will enhance the self-absorption effect of Al and will reduce the self-absorption effect of Cu and Si compared to other noble gases, which is analyzed for the following reasons: 1. Argon has a smaller thermal conductivity and a higher atomic mass compared to air, whereas He is a relatively light ambient gas in which the plasma is free to expand without any significant confinement effect, and Ar is a heavy gas, which leads to a lower velocity and more confinement effects in the plasma plume. 2. During laser ablation processes in gases with heavier atomic weights, such as argon, the vapor plume experiences a reduced expansion rate due to the greater momentum drag imposed by the surrounding gas. Additionally, the plasma's inertia increases in these denser gases, causing it to expand more slowly compared to lighter gases. Consequently, the plasma maintains a higher temperature for extended periods, giving rise to intense plasma emissions. This elevated temperature and emission intensity lead to a higher electron density within the plasma. 3. Ambient gases significantly influence the chemical processes during laser ablation, including the generation of oxides within the laser-induced plasma. By employing argon as the surrounding atmosphere, the excited atoms are safeguarded from undergoing oxidation, thereby mitigating the formation of oxides. This protective effect diminishes plasma emission intensity and shortens its lifespan. 4. This is because, due to its higher inertia, the plasma expansion in heavy gases is slower than in light gases, keeping the plasma temperature high for longer periods of time and generating strong plasma emissions. 5. Argon is easier to ionize and produce electrons, and the sample under test is capable of generating a larger electron density, high plasma temperature, and strong spectral signals, compared with air is more difficult to ionize. In the argon atmosphere, the intensity of the background spectral line is also improved; this is due to the electron density and plasma temperature in the argon atmosphere, and the decay rate is lower than in the air atmosphere.

Through the above researcher's experimental analysis, it is found that Ar has an enhanced effect on the self-absorption of certain elements' spectral lines, but Ar can also enhance the spectral signals. However, the composition of the medium is complex and contains the elements it discussed above, which may enhance the self-absorption of the spectral lines between their elements in Ar, and there is a risk of increased self-absorption of certain elements' spectra if Ar is used to enhance the signals in the detection medium.

In summary, LIBS in mordant for self-absorption effect is less studied, mainly because the composition of mordant is too complex, the self-absorption effect between components is too complicated, while the method to reduce the self-absorption effect may vary depending on the specific LIBS detection system and coal sample characteristics. However, it is possible to improve the overall accuracy of mordant analysis by learning the application of LIBS in the detection of other samples rather than focusing on a specific element.

(3) Matrix Effect

Essentially, matrix effects in LIBS stem from the disparities in physical and chemical characteristics between standards and samples. These disparities encompass the sample's composition, melting point, volatility, moisture content, reflectivity, and surface texture, all of which contribute to the observed variations [63]. Variations in physical and chemical attributes lead to distinct laser ablation behaviors and interactions between the plasma and

ambient gas, ultimately yielding disparate plasma compositions and spectral profiles. In the context of coal analysis, the intricate sample matrix encapsulates an extensive range of elements, nearly mirroring the composition of Earth's crust. Different physical and chemical properties of coal will lead to nonlinear laser–matter interactions during laser ablation, plasma formation, plasma expansion, etc., which will result in different plasma properties for different samples, thus affecting the accuracy of the measurement results [31]. Often, physical matrix effects can be largely eliminated by simple pre-treatment (e.g., pressing pulverized coal into briquettes) [64]. However, matrix effects due to chemical properties are the main challenge for quantitative analysis of coal composition by LIBS.

The different chemical forms of the elements have a certain influence on the intensity of their characteristic spectral lines. By studying the spectral properties of soluble starch, anhydrous p-aminobenzenesulfonic acid, calcium carbonate, and graphite under different laser energies, it can be concluded that the more complex the structure and the larger the chemical bonding energy of the existent form of the C element is, the larger the laser energy needed to be excited; and due to the differences in the chemical compositions and the size of the atomic force within the molecule, the spectral properties are different. Moreover, due to the differences in chemical composition and the magnitude of intra-molecular atomic forces, the spectral properties of different forms of C elements vary greatly [65].

The chemical composition varies greatly among different types of coals, and from the above, it can be seen that the presence of the C element has a great influence on the laser plasma properties of coals. The media are usually categorized into lignite, bituminous, and anthracite. Among them, lignite is the most volatile coal, bituminous coal is the second, and anthracite is the least; the ash content of anthracite is the most, bituminous coal is the second, and lignite is the least, and there are large differences in their chemical compositions. By comparing the spectral signals and plasma properties of these three kinds of coals, it is found that the ash in coal mainly consists of solid compounds of elements such as Si and Al, and the more the ash, the more the content of Si and Al elements, so the intensity of Si and Al elements is the largest in anthracite, followed by bituminous coal, and the smallest in lignite. In addition, because the volatile fraction in coal is mainly composed of gases containing C, H, O, and N, the ionization energy of C, H, O, and N elements is higher than that of the ash elements such as Si and Al, which leads to the fact that the coal samples with low volatile fraction and high ash content are more likely to be pierced by ionization and ablated more under the same laser energy, and the plasma temperature and electron density are the largest in anthracite, the second largest in bituminous coal, and the smallest in lignite [66].

Strategies aimed at mitigating chemical matrix effects can be broadly categorized into three groups. One intuitive approach involves the meticulous synthesis of materials to create matrix-matched standards. A widely employed technique is to blend ground or molten samples with a substantial amount of identical matrix material, thereby making it the primary constituent [67]. Another viable treatment method involves dissolving and evenly spreading each sample onto a metal substrate, resulting in the formation of a thin layer [68]. Here, by ablating the metal substrate prior to the sample, the material from the substrate predominates in the plasma, thereby reducing matrix effects. The second tactic, known as internal calibration, entails incorporating an internal standard into each sample. This allows for the normalization of the analyte signal relative to the internal standard signal, enhancing the accuracy and precision of the analysis [69]. This method necessitates the careful selection of an appropriate internal standard element that mimics the behavior of the analyte, exhibits uniform distribution, and produces a discernible signal. Additionally, a third approach involves developing calibration methods that directly assess matrix effects or the inherent characteristics of the matrix. Several investigations have documented the normalization of ablation mass spectra, achieved by quantifying indirect signals such as the dimensions of the ablation crater or the acoustic energy released during the process [70].

For quantitative analysis of coal composition using LIBS, the above three methods usually choose the internal standard method to reduce the influence of the matrix effect. Gu also proposed a new method, the adaptive subset matching (ASM) method, to reduce the influence of the matrix effect. The proposed method underwent rigorous evaluation using 90 coal samples to ascertain its carbon content determination capabilities. The findings revealed that the application of ASM significantly enhanced the quantitative accuracy of both multiple linear regression (MLR) and partial least squares regression (PLSR). Specifically, the root-mean-square-error of prediction (RMSEP) for MLR diminished from 6.19% to 3.23%, while that of PLSR decreased from 2.83% to 1.59%. Concurrently, the average inter-pulse relative standard deviations (RSDs) predicted by MLR and PLSR were reduced from 13.8% to 8.43% and 4.59% to 2.48%, respectively. Furthermore, notable improvements were observed in the quantitative analysis of calorific value, nitrogen, and hydrogen content. These outcomes underscore the efficacy of ASM in mitigating matrix effects during coal analysis via LIBS. In contrast to models reliant solely on calibration samples, ASM incorporates information specific to the sample matrix, enabling effective correction of matrix effects while preserving the simplicity and swiftness of LIBS measurements. However, it is crucial to incorporate a sufficient number of samples into the ASM method, making it particularly advantageous for industrial settings characterized by abundant data generation [63].

3.2. Sample Section

(1) Pressure Effect

For pulverized coal samples, the structure is loose, and the surface is irregular, which will splash if directly excited by the laser and is unfavorable to the collection of spectra, so it is necessary to press the coal powder into a coal cake with better mechanical properties by means of a press. If the pressure of the ablated material is too small, the particles can easily escape into the atmosphere, thus intercepting the laser beam and affecting the formation of plasma [71]. On the other hand, when the laser is applied to the surface of the sample, there is a strong collision with the ambient air that generates a shock wave, so the laser is applied to surfaces with different densities and produces craters with widely varying properties. Compaction produces a uniform, smooth, analyzed surface exposed to the laser beam. In pressing pulverized coal, the pressing force should be high enough, and the pressing time should be long enough. When the pressing force or pressing time is insufficient, the spectral effect also deteriorates. The intensity of the spectral line is related to the ablation amount of the sample, which depends on the number of particles of the sample in the ablation area and the ablation depth. Under the same experimental conditions, the focusing spot of different densities of coal samples is basically the same, and the density of powdered coal samples is smaller than that of flake coal samples. With the increase in pressure, the density of the flake samples increased slightly, but the change was not significant. The powdered coal samples have fewer particles interacting with the laser in the focusing spot, but the ablation depth of the powdered coal samples is larger than that of the flake coal samples due to the looseness of the powdered coal samples and the ablation depth of the coal samples has a larger impact on the spectral line intensity in the process, but the impact of the ablation depth of the samples is relatively small with the increase in the pressure [72].

We measured the effect of pressure on the composition of the LIBS detection medium by taking sample coal from Hebei province as an example. At laser energy of 90 mJ, sample pressures of 10 MPa, 20 MPa, 30 MPa, and 40 MPa were selected as variables [73], and parallel experiments were conducted, resulting in a minimum average RSD for samples at 30 MPa, which is because the pulse energy is coupled with more substances when the pressure of the press is increased, and the ablated mass is no longer increased when the pressure is more than 30 MPa [74].

(2) Pulverized Coal Particle Size

Coal dust particle size, or the diameter of coal dust particles, is the aperture of the smallest sieve hole through which coal dust can pass under a certain vibration intensity and sieving time. The uniformity of coal particle size has a great impact on the quality of LIBS spectra. If the distribution of coal particle size is not uniform, then during the laser excitation process, coal particles with different particle sizes will produce different spectral signals, resulting in spectral overlap and interference, diminishing the precision and dependability of LIBS assessment. The size of the coal particle size also directly affects the interaction between the laser and the coal powder, and a smaller coal particle size can provide a larger surface area so that the laser energy can be more easily absorbed and converted into plasma, thus improving the excitation efficiency of LIBS. Li et al. analyzed coal particles from four different size groups (0–0.1, 0.1–0.2, 0.2–0.3, and 0.3–0.6 mm) and compared them to coal particles deposited on filters and analyzed with LIBS, and it was found that the RSDs for particle sizes of 0–0.1 or 0.3–0.6 mm were small, which may be influenced by two factors: one is the intermolecular forces and the other is the properties of the surface. For smaller particle sizes, the intermolecular forces between particles are larger and the pelletized coal samples are rigid and therefore plasma stable. For larger particle sizes, the individual particles possess a greater surface area, with the majority of laser energy being focused primarily on the exterior surface rather than penetrating the interstitial spaces between them. However, pelletized samples made from coal with large particle sizes are friable. Therefore, small particle-size coals are usually chosen for measurements [75].

(3) Binder

When LIBS is used to inspect powdered materials such as anthracite samples, the anthracite is poorly pressed into tablets due to its high degree of coalification, high carbon content, high calorific value, low volatile matter, high hardness, high mechanical strength, and poor abradability. It either does not press into flakes, or the surface of the pressed particles is not tough enough to withstand multiple laser shots. As a result, the laser interaction with the anthracite varies too much from pulse to pulse due to sudden changes in the sample surface, leading to unstable ablation quality and spectral signals [76]. Therefore, an additional binder is usually added to improve the compressibility of the sample and the mechanical strength after pressing [77].

When anthracite is not used as a binder to make the compact, the particle's compact surface is easily delaminated during LIBS measurements when the laser pulse is directed to its surface. This leads to a large measurement uncertainty because the laser–surface interaction can be very dynamic due to the exfoliation of the laser pulse surface. Eleven samples were used for the experiments with the highest RSD value of 59.9% and an average RSD value of 30.7% for eleven samples, and the results were not reproducible [67].

However, the selection of the appropriate binder is paramount to ensuring accurate final measurement outcomes. Ideally, this binder should effectively unite the anthracite powder while exhibiting a distinctly different elemental makeup from the anthracite itself, thereby eliminating potential interference. Nevertheless, the multitude of elements present in anthracite poses a significant challenge in identifying an ideal binder that meets these criteria. However, anthracite contains only a small percentage of elements other than C, H, O, and N. If the binder is mixed with anthracite in the proper proportions, the binder will contain a dominant amount of these inorganic elements, while the corresponding primitive elements in the anthracite will be negligible. For the spectral selection of the anthracite samples, only two carbon feature lines are not overlapped and have a large enough signal-to-noise ratio at C(I) 193.09 nm and C(I) 247.86 nm. The spectrum centered at C(I) 247.86 nm is more symmetrical, so the carbon feature line at C(I) 247.86 nm is selected to prove the final measurement precision and accuracy [67]. Zhu et al. used KBr to polymerize coal or charcoal into pellets and demonstrated that a high content of KBr provides a good environment for the formation of stable emitting plasma with relatively high electronic

excitation temperature and electron density. The multivariate linear regression (MLR) model and the artificial neural network (ANN) model were used for the prediction of KBr-containing mordant samples with a coefficient of determination (R^2) of 0.977 and 0.991. The reason is that KBr inhibits the pyrolysis of volatiles induced by laser ablation [78].

Since the binder can be easily controlled to a certain percentage, some fixed concentrations can be automatically added and used as internal calibration elements. Liu and Lu also demonstrated that Si is an effective internal calibration element for the measurement of C by simulating a coal matrix with graphite and SiO_2 added by the analog compound $C_6H_6NO_3$ [79]. $Na_2SiO_3 \cdot 9H_2O$ was found to be a suitable binder for anthracite samples with an optimum weight percentage of 50%. The suitability of the binder for anthracite carbon measurements was demonstrated by the reduction in the average RSD value of the internal calibration method from the original intensity mean of 30.7% to 12.1%, with an RMSEP of only 6.25% and an average relative error of 10.14% for the eleven samples selected. It can be clearly seen that the RSD of the samples with a binder is greatly reduced, indicating a great improvement in the application of binders in the quantitative analysis of anthracite coal [67].

According to the above analysis, LIBS can be used to measure fly ash and other powdery materials, and the selection of the binder not only requires good adhesion and does not interfere with the analyzed elements but also can be used as a standard accession as an internal standard for the analyzed elements. The comparison of the LIBS measurement of unburned carbon after mixing the two binders KNO_3 and Na_2SiO_3 with fly ash samples shows that for the measurement of unburned carbon in fly ash, the use of Na_2SiO_3 as a binder not only improves the compressibility of the fly ash but also its Si can be used as an internal standard in the analysis of unburned carbon, which can further improve the accuracy of quantitative analysis and improve the accuracy of the quantitative analysis of unburned carbon and improve the detection limit. In addition, using Si 251.61 nm as the internal standard under the same conditions can help to improve the reliability and sensitivity of the unburned carbon measurement of fly ash [80].

3.3. Experimental Section

(1) Laser Energy Optimization

The laser energy directly affects the spectral intensity value, the laser energy is small and cannot penetrate the sample, and the laser energy is too large to make the plasma have a shielding effect. Generally speaking, the relative standard deviation (RSD) with the laser energy becomes larger with the law is first smaller and then larger; the reason for this change process is that with the increase in laser energy, the number of excited particles increases, but when the energy increases over a certain threshold, the spectral line is prone to the self-absorption effect [32]. According to some research results of LIBS technology used for coal quality analysis, it was found that most of the laser energies for excitation of coal were concentrated at 70–110 mJ, so we carried out some research by taking the sample coal from Hebei province as an example. The RSDs of the main spectral lines at laser energies of 70, 80, 90, 100, and 110 mJ were solved under the working conditions of the focus buried 2 mm into the sample, sample pressure of 30 MPa, and delay time of 1.2 μs. The results show that the laser energy has the least effect on the RSD of the spectra when the energy is 90 mJ, and this parameter was selected for the subsequent experiments.

(2) Focus Position

In LIBS, focused laser pulses are used to ablate material from a surface and form a laser plasma that excites the vaporized material. The focal point is generated at one of three locations: above the sample, at the surface, or inside. When the position of the focus is above the surface, air breakdown always occurs, and the generated spectra are unstable and unreliable, so they are not considered. The focusing lens to the surface distance (LTSD), or the vertical distance between the laser focus and the sample surface when it is not exactly on the sample surface as shown in Figure 3 (taking into account that the focus may be

slightly embedded in or away from the sample surface), directly affects the size of the ablation crater formed by the laser on the coal cake sample. This variation in distance affects the energy distribution and density of the laser at the ablation point since the degree of focusing of the laser energy and the energy density are key factors in the interaction of the laser with matter.

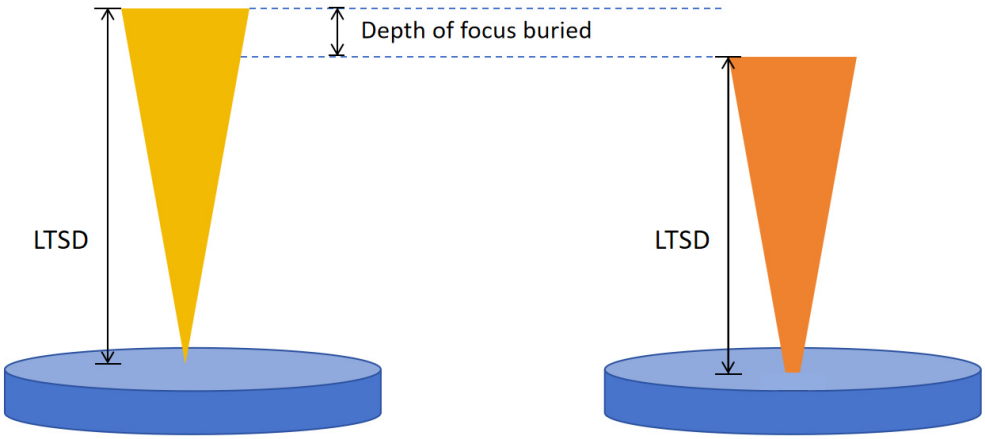

Figure 3. Schematic of the focal depth of the lens buried.

We investigated the effect of the laser focus position on the composition of the LIBS detection medium using a sample coal from Hebei Province as an example. Under the laser energy of 90 mJ, delay time of 1.2 μs, and sample pressure of 30 MPa, the working conditions of focal point buried depths of 0 mm, 1 mm, 2 mm, and 3 mm were selected, and the changes in RSD of the main spectral lines were compared. When the focal point is located inside the material, the error shows a tendency to decrease and then increase with the increase in depth, and the average RSD of the main spectral lines is the smallest when the focal point is buried 2 mm into the surface of the sample, which may be due to the fact that the plasma generated on the surface is flatter and relatively cooler [75].

(3) Static and Dynamic Analysis

In order to analyze the composition of the coal cake more comprehensively and to ensure the accuracy and reliability of the LIBS test, we should avoid continuously focusing the laser as the same location of the coal cake during the LIBS test. This is because prolonged focusing may result in high elemental concentrations or the introduction of impurities in the area, thus distorting the true assessment of the overall composition of the coal cake. In order to avoid this problem, we usually use a multi-point measurement method to more accurately characterize the overall composition of the coal cake by performing LIBS analysis on multiple points at different locations and averaging the results. This method can provide more comprehensive and balanced data, thus ensuring that our judgment of the composition of the coal cake is more accurate and reliable. The analysis compares static multi-point and dynamic continuous measurements in which the dynamic analysis is performed by delivering 30 laser pulses at 10 Hz, and the sample is moved at 2.5 mms^{-1}; on the other hand, the static analysis is performed by panning the coal cake after delivering 12 laser pulses deeply at the same sample position. M.P. Mateo et al. analyzed the dynamic analysis of the coal cake composition of eight coal mines in two working modes by comparing the dynamic and static modes of operation of the Nd: YAG laser at 1064 nm and 355 nm in eight coal mine samples for some of the main inorganic species Fe, Mg, Si, Al, and Ca in coal. The emission spectral lines for which no self-absorption or interference effects with other elements were observed were selected: Fe(II) 274.948 nm, Mg(I) 285.213 nm, Si(I) 288.158 nm, Al(I) 309.271 nm, and Ca(II) 315.887 nm. Among them, the static mode of analysis reveals a better RSD result because, in dynamic mode, the static mode of analysis is usually chosen for LIBS because of the presence of surface contamination effects in the dynamic mode [81].

(4) Number of Laser Actions at Focal Position

Upon focusing a high-energy laser pulse onto the sample, the laser energy is concentrated within a confined region, causing the local temperature to soar swiftly beyond the vaporization threshold. This results in the generation of a laser-induced spark on the sample's surface, where the material undergoes ablation and subsequent atomization within the laser-induced plasma. For a pressed coal cake, the number of laser actions at the focal position has a significant effect on the acquisition of spectra. When 120 laser pulses are struck on the surface of a sample, the test is performed in the following three modes: In the first mode, a total of 120 laser pulses are directed at a single, fixed position on the sample. In the second mode, a spectrum is generated by firing 12 laser pulses at the same sample location, and this process is repeated across 10 distinct positions. In the third mode, each laser pulse strikes a unique, unexposed position on the sample, ensuring that each hit exposes a fresh area.

The RSD is relatively large when the sample is in the first and third modes. When the sample is in the first mode, this is due to the craters generated by the laser pulse. In this case, the size of the craters increases over time, affecting not only the sample ablation rate but also the way the plasma is imaged on the fiber [75]. And through daily experiments, it was found that the third type of spectral error is the largest and the most unstable spectrum because its focus is located below the surface of the sample; the first hit will penetrate the surface of the material, resulting in the evaporation of the pulverized coal, which will lead to the focus of the laser is different from the subsequent laser pulses, which will affect the optical fiber light harvesting, and the laser pulses will be absorbed in front of the sample, which will lead to a very poor RSD [82]. It also suppresses signals from the surface that may have been affected by contamination during preparation [81].

(5) Ambient Gas

Ambient gases play a pivotal role in the formation of plasma and can significantly influence the effectiveness of LIBS analysis. For this investigation, various ambient gases, including air, nitrogen, and argon, were employed to evaluate their impact. The analysis yielded Mg(II) 279.6 nm, Mg(I) 285.2 nm, Ca(II) 393.4 nm, and Ca(I) 422.7 nm spectral line intensities moderately elevated in air compared to argon. However, the analysis revealed that argon yielded significantly greater line intensities, with a notable increase of approximately three- to fourfold compared to those observed in air or nitrogen. It is remarkable to mention that the spectral line intensity of O(I) 777.4 nm is approximately four times more intense in air than in nitrogen. Air is predominantly composed of nitrogen (roughly 78% by volume) and oxygen (around 21% by volume), accompanied by trace amounts of other gases. The presence of O(I) 777.4 nm spectral lines in air can be attributed to both the oxygen content inherently present in the atmosphere and that emanating from coal or other elemental sources of oxygen [75].

Alternatively, a suitable gas can be introduced to encompass the ablation zone, serving as a shield for the target area. This approach safeguards the ablation region from potential harm caused by unwanted environmental and atmospheric contaminants. Furthermore, it effectively eliminates coal particles or other debris that may detach from the sample during the laser ablation process, thus preventing these particles from disrupting the incoming laser pulse and plasma emission. The results of the analysis yielded improved atomic (Cl, Si, Fe, Mg, Al, Ca, Na, and K) and molecular (CN and C_2) emissions from the argon gas stream and improved molecular carbon emissions (CN and C_2) from the N_2 gas stream. The flow rate of any gas except argon does not affect the atomic and molecular C emission. Due to the Ar-induced plasma shielding effect, C emission is reduced by increasing the Ar gas flow rate. The Ar environment leads to plasma confinement and re-excitation of plasma material due to larger atomic weights and densities. As a result, the optical signal intensity, plasma temperature, and electron density are higher than in the N_2, air, and helium environments [83].

(6) Delay Time

Due to the tough radiation properties of plasma, the spectral intensity changes with time from weak to strong and then weak, while the effect of continuous radiation can be reduced to improve the signal-to-noise ratio, so the timing relationship between the laser and the spectrometer needs to be set up, and this time is the delay time. When the laser is emitted, a microsecond delay time triggers the spectrometer to collect the LIBS spectra, and a reasonable delay time can minimize the signal-to-noise ratio and RSD of the spectra. In the case of the same elements and the same characteristic spectral lines, the optimal delay time of the spectral lines may be different for different coal samples. This is related to the characteristics of the coal samples, such as moisture, volatile matter, and ash content, which will lead to a reduction in the accuracy of the simultaneous measurement of major and minor elements in coal [84]. However, it has been shown that the problem can be alleviated to some extent by choosing a larger sampling gate width [85]. However, the delay time is not without pattern, and by qualitatively analyzing the characteristic spectral lines of the elements, the time evolution law of the signal-to-noise ratio of the characteristic spectral lines was calculated.

Taking the sample coal from Hebei Province as an example, we selected delay times of 0.8 μs, 1.0 μs, 1.2 μs, 1.4 μs, and 1.6 μs as the variation parameters and carried out parallel experiments at a laser energy of 90 mJ, a focal point burying depth of 2 mm and a sample pressure of 30 MPa. The results show that the average RSD of the main spectral lines is minimized when the delay time is 1.2 μs. This shows that the signal-to-noise ratio undergoes a process from small to large and then decreases with increasing delay time [84].

After the above summary, this section explores the application of LIBS technology in media analysis, analyzes in depth the possible sources of errors at the experimental and principle levels, and discusses strategies and methods to improve the measurement accuracy. Among them, the self-absorption effect and matrix effect of spectral lines as the key factors affecting the measurement accuracy; although they cannot be completely eradicated, we can effectively weaken their influence through a series of methods, and there is still room for further research and optimization in this field. As for other parameter settings in LIBS experiments, the optimal experimental parameters often vary from case to case due to different experimental equipment and detection media; therefore, this section aims to provide a framework and ideas to enhance the measurement accuracy of LIBS in media analysis, rather than a set-in-stone guideline, and researchers should make flexible parameter adjustments and optimizations according to their own experimental conditions.

After ensuring the rationality of LIBS spectral acquisition, through reasonable parameter settings and accurate spectral acquisition, we are able to obtain more accurate and reliable raw data, and then through advanced data processing techniques and algorithms, we can further improve the precision and accuracy of the media analysis, which lays the foundation for the next section: the data processing of LIBS technology.

4. Spectral Data Processing Models and Their Advantages and Disadvantages

From the above, it can be seen that the LIBS technique, like other analytical techniques, has a matrix effect, i.e., the physicochemical composition of the sample affects the emission spectral signals generated during laser ablation of the sample, and although the matrix effect can be reduced by various experimental methods, it cannot be completely eliminated, and on the other hand, it also indicates that there is a great correlation between the characteristics of the spectrum and structural properties of the sample [79]. The quantitative effect of the LIBS technique is poor, and the mechanism of laser interaction with different substances is not yet very clear, resulting in the lack of a strictly physical model for the calibration of elemental content. For the measurement of solid samples with matrix differences, the LIBS technique has certain matrix effects that lead to the loss of one-to-one correspondence between spectral intensity and elemental concentration, and therefore, specific calibration algorithms need to be proposed to solve the problem of quantitative measurement. Based on this feature, various analytical methods are introduced to obtain the LIBS spectral

information related to the elements so as to establish the correlation formula between the spectral information and the volatile fraction.

4.1. Linear Solution

(1) Principal component regression

Principal component regression (PCR) is a regression analysis method that uses the principal components obtained from principal component analysis (PCA) as independent variables to predict a dependent variable. It first performs PCA analysis to obtain the principal components. Subsequently, these principal components are harnessed as the independent variables to generate predictions within a regression framework. Here, PCA, a prevalent multivariate statistical technique introduced by Karl Pearson in 1901, plays a pivotal role [86]; it is an unsupervised learning method which removes redundant information from a multidimensional dataset by pattern reduction [87]. PCA not only optimizes the integration of multivariate data information, extracting a reduced set of comprehensive variable features that encapsulate the majority of the original data's information, but it also diminishes the dimensionality of high-dimensional data spaces by adhering to the principle of minimizing the loss of data information. The essence of PCA is to perform a linear combination of each variable in the measurement data matrix, and after dimensionality reduction, we can obtain the new variables that contain the information of the original data, and these new variables are the principal components in PCA. Analyzed from a mathematical geometric point of view, PCA is to find a projection direction so that the projection value of all data points in that direction is the largest, which is equivalent to the largest variance in statistics, and its chemical meaning is the largest amount of information retained.

PCA is based on the premise of minimal loss of information; the original variables are synthesized into fewer principal components through linear transformation. Usually, its principal components have the following characteristics: (1) The count of principal components is substantially lower than the total number of original variables. After the original variables are synthesized into a few factors, the factors will be able to replace the original variables to participate in data modeling, which will greatly reduce the computational workload in the analysis process. (2) Principal components can reflect most of the original variables. The information factor is not a simple rounding off of the original variables but the result of the reorganization of the original variables, so it will not cause a large amount of loss of information on the original variables and can represent the vast majority of the information on the original variables. (3) The principal components should be unrelated to each other. The new principal components derived from principal component analysis are not related to each other, and the participation of factors in data modeling can effectively solve the problems brought by overlapping variable information, multicollinearity, and so on. The quality of the PCA model can be assessed by using cross-validation [88]; however, in the process of dimensionality reduction in PCA, the dependent variable is not involved in guiding the construction of the principal components, so the PCA does not guarantee that the direction of the predictor variable is well explained, while the direction of the dependent variable is not well explained by the PCA, while the dependent variable is predicted satisfactorily.

When using LIBS for media analysis, due to the complexity of the composition of the media, the amount of data information obtained in the spectral analysis is very large, and these variables reflect certain characteristics of the media samples in different degrees, and this information may overlap to some extent. Then, principal component analysis can be used to extract the principal components that are representative enough of the original spectral data in order to achieve the goal of reducing the number of variables and simplifying the analysis of the complexity of the data can be simplified by reducing the number of variables and analyzing the data.

Liu et al. chose 16 samples of data as the corrected data and set up a correction model to determine the minimum RMSECV value when the main component was 14. The cumulative explanatory variance was consistent with the 90% cumulative explained

variance [89], and the measured results were 0.9986, 0.9891, 0.9958, and 0.9979 for ash, fixed carbon, volatile matter, and calorific value R^2, and 0.2455%, 0.662%, 0.6857%, and 0.098% for RMSECV [4].

Zhang et al. determined 40 standard coal samples using the principal component regression model, and a total of 25 samples were selected as calibration samples, while the remaining 15 samples were used as validation samples for prediction. For the predicted samples, the mean absolute errors (MAE) for the determination of their ash, volatile matter, and calorific value were 1.4, 0.94, and 0.91, the RMSEPs were 1.636, 1.149, and 1.126, and the average relative standard deviations (ARSD) were 2.26, 1.67, and 0.32 [90].

Today, in the context of rapid development of artificial intelligence, machine learning algorithms have made significant progress. However, in the scenarios where PCR is applied to LIBS technology, PCR does not show significant advantages in terms of both data processing speed and accurate determination of results. Currently, the research focus of PCR has shifted to the precise differentiation of coal quality differences, especially the analysis of differences in elemental composition in coal from different mining areas, with a view to improving coal combustion efficiency and reducing pollution emissions.

(2) Partial least squares method

Partial least squares regression (PLSR) is a multivariate statistical technique rooted in factor analysis, adept at efficiently disentangling and mitigating the complexities of multiple correlations within datasets [90]. PLS is a supervised method; it is similar to PCA in that the original spectral data are linearly combined and downscaled to obtain a new set of variables, where the new variables obtained are called PLS factors. After that, these PLS factors are then utilized as new variables in a multiple linear regression. The new variables maintain as much information as possible about the variance of the original variables while also having a strong explanatory power for the dependent variable [91]. PLS achieves two significant objectives: it compresses the information of the original variables into a set of extracted feature variables that adequately encapsulate their essence and concurrently possesses remarkable explanatory capacity towards the dependent variable. This method adeptly constructs mutually orthogonal feature vectors for both independent and dependent variables by projecting their high-dimensional data spaces onto corresponding lower-dimensional subspaces. Subsequently, it establishes straightforward linear relationships between these feature vectors, thereby transcending the challenges posed by covariance issues. During the eigenvector selection process, PLS underscores the explanatory and predictive prowess of independent variables over the dependent variable, efficiently purging unhelpful noise from the regression model. This ensures that the model remains parsimonious, containing only the most essential variables [92–94].

Zhu et al. utilized LIBS to reconstruct the principal components (PC), elucidating the interplay between variance and calibration values via comprehensive spectral wave point analysis. This approach yielded an enriched set of components intimately tied to the characteristics of interest, thereby optimizing the predictive power. Subsequently, they embarked on quantitative modeling of ash content, volatile matter, and calorific value, adopting partial least squares regression (PLSR) as an advanced alternative to ordinary least squares (OLS), leveraging its strength in handling complex datasets and enhancing model accuracy. The R^2 training and test sets for ash and volatile fractions improved from 0.8125 and 0.8222, 0.6502 and 0.6483 to 0.9701 and 0.9818, and 0.9458 and 0.9429, respectively. The cross-validation root-mean-square error of coefficient of variation (RMSECV) and RMSEP improved from 1.7843% and 1.7696%, 1.1797% and 1.4301% to 0.7153% and 0.7037%, 0.5678% and 0.6628%. The calorific value R^2 was also improved from 0.7086 and 0.7157 to 0.9857 and 0.9811 for the training and test sets. RMSECV and RMSEP were reduced from 0.9525 MJ/kg and 1.0178 MJ/kg to 0.1518 MJ/kg and 0.1613 MJ/kg, respectively [1].

Zhang et al. utilized the PLSR model on a dataset comprising 40 standard coal samples. Out of these, a strategic selection of 30 samples served as the calibration set, while the remaining 10 samples were meticulously reserved for validation purposes, offering a robust

means to assess the predictive performance of the model. For the predicted samples, the MAs of ash, volatile matter, and calorific value were determined to be 1.27, 0.82, and 1.02, the RMSEPs were 1.634, 1.058, and 1.331, and the ARSDs were 1.8, 1.59, and 0.48 [90].

Zhang et al. used a support vector machine (SVM) optimized by a genetic algorithm (GA) to classify coal according to ash content, and PLSR was used to build different models for each coal type. The test sets of R^2, RMSECV, and RMSEP for ash were 0.9945, 0.6313%, and 0.9065%, respectively. The R^2, RMSECV, and RMSEP for the test set of volatile matter were 0.9888, 0.4989%, and 0.7719%, respectively. The R^2, RMSECV, and RMSEP for the test set of calorific values were 0.9969, 0.9906, 0.2331 MJ/kg, and 0.4093 MJ/kg, respectively. The results were improved for the models built without classification [5].

4.2. Machine Learning

LIBS spectral analysis is known to be very complex, with many complex interactions at play [95]. Standard approaches like univariate analysis and multiple linear regression (MLR) concentrate solely on elucidating the link between analyte concentration and prominent spectral features. However, to elevate the quantitative accuracy of LIBS, adopting sophisticated multivariate linear models such as principal component regression (PCR) and partial least squares regression (PLSR) proves instrumental. These methods adeptly mine the full spectral data, enabling the extraction of comprehensive elemental information that surpasses the capabilities of simpler models [96]. However, the performance of these models often encounters diminishing returns when confronted with intricate and nonlinear relationships between spectral intensity and concentration [97]. Machine learning stems from the advancements in artificial intelligence, furnishing computers with the capability to discern patterns and connections within intricate datasets that might otherwise remain elusive or undetected by conventional analytical means. By merging insightful yet intricate data with analytics tailored to delve into highly complex data realms, it uncovers meaningful insights that transcend traditional methodologies [95]. Machine learning is a multi-disciplinary cross-discipline that encompasses many learning algorithms. These algorithms cover a wide range of aspects, from simple linear regression to complex deep learning networks. The following is a summary of algorithms applicable to LIBS analysis of mordant elements in recent years, which is intended to provide references and insights for subsequent research:

(1) Support vector regression

Support vector regression is a regression algorithm with good generalization performance based on the regression method of structural risk minimization idea. It can solve the regression problem for a given dataset $D = \{(x_i, y_i)\}_{i=1}^{N}$ obtained from the latent function, where x_i denotes the sample vector, y_i denotes the corresponding response, and N is the total number of samples [98]. The principle is to first map the original data nonlinearly into a high-dimensional feature space and then fit a linear function to approximate the potential function between x and y [98], or to find a fitted curve from which the distance to the data points can be minimized [99].

During LIBS detection of mordant samples, SVR can map the spectral data to a higher dimensional space by using a nonlinear kernel function. Then, the correlation function between the mapped spectral data and the coal quality is approximated and fitted [100]. Zhang et al. used the SVR model for 40 standard coal samples, and a total of 30 samples were selected as calibration samples and the remaining 10 samples were used as validation samples for prediction. For the predicted samples, the MAEs of ash, volatile matter, and calorific value were determined to be 1.13, 0.86, and 0.66, the RMSEPs were 1.316, 1.066, and 0.92, and the ARSDs were 1.69, 1.33, and 0.18 [90].

(2) Random forest

Random forests are a class of algorithms for solving classification and regression problems with strong resistance to overfitting and model generalization [101]. As integration methods, they grow several trees as base estimates and aggregate them for prediction,

which can be constructed by randomizing the feature set, the dataset, or both [102]. Random forests can handle very high-dimensional data and nonlinear relationships between predictor variables [99], and their generation process consists of one or more categorical regression trees (CARTs) with the following main process: The model randomly selects a number of features and generates tree nodes with variance. When the nodes can no longer be split, decision trees will be formed, and the construction of the forest with multiple decision trees will be repeated. Finally, the classification results are derived from the voting results of all decision trees [34]. The average of the prediction results will be considered as the final prediction result [103]. The RF model is constructed according to the above process, which allows for nonlinear fitting between spectral data and elemental compositions in the LIBS detection medium.

Zhang et al. advocated that LIBS technology, when bolstered by diverse machine learning models, can precisely and efficiently ascertain the content of non-metallic elements within coal. They employed linear regression, support vector regression, and random forest algorithms as calibration and predictive tools, showcasing the versatility and accuracy of this approach, with R^2 values of 0.9844, 0.9625, and 0.9829 for C, H, and N, respectively, and root-mean-square errors (RMSE) of 0.2669, 0.2508, and 0.0335, respectively [103].

(3) Kernel extreme learning machine

Kernel extreme learning machine (KELM) stands as a highly esteemed nonlinear, multivariate calibration technique renowned for its swift learning capabilities and remarkable generalization prowess. It adeptly circumvents the common challenges associated with traditional neural networks, such as sluggish training processes and the risk of overfitting. At its core, KELM incorporates a kernel function into the extreme learning machine (ELM) framework, supplanting the random mapping inherent in basic ELM with a kernel-based mapping strategy. This innovation significantly enhances the generalization capability and stability that might otherwise be hindered by the arbitrary assignment of hidden layer parameters. Furthermore, KELM substantially diminishes computational complexity, obviates the need for intricate optimization of the hidden layer node count, and efficiently yields an optimal least squares solution, all while maintaining a concise exposition [104]. The process is as follows: Firstly, choose a suitable kernel function to map the input data into a high-dimensional feature space. Secondly, the similarity between the training data is calculated by the kernel function, and the kernel matrix is constructed. Finally, map the input data to the hidden layer space and generate a new feature representation, calculate the linear relationship between the hidden layer output and the target output, and obtain the output weights. The method is applicable to the fitting of spectral data and elemental components in LIBS.

Yan et al. used V-WSP-PSO to select the number of KELM features and construct a KELM model to fit the LIBS detection medium spectra to determine the affinity value. The process is as follows: Firstly, the V-WSP method is used to eliminate irrelevant and redundant features to form a simplified input subset, and then the PSO method is used to further refine the retained features to find a small number of features with high prediction accuracy. Finally, a calibration model is established based on the nonlinear multivariate calibration method of KELM. In the quantitative assessment, a total of 28 samples were randomly allocated, with 20 samples designated for the calibration set and the remaining 8 samples reserved for the prediction set. The results show that the prediction result of calorific value is 0.9894 for R^2 and 0.3534 MJ/kg for RMSEP [105].

(4) Synergistic regression

Song et al. introduced a novel model named synergistic regression (SR), tailored specifically for the fitting of spectral data with elemental composition. This model harmoniously integrates nonlinear and linear modeling techniques for data regression, thereby preserving the precision of nonlinear methods while concurrently elucidating the individual contributions of specific variables to the prediction outcomes. The main process is as follows: Firstly, the input matrix undergoes a transformation into a kernel matrix form, facilitating

nonlinear regression through low-dimensional projection. Subsequently, a penalty term is incorporated to regularize the regression coefficients within the kernel matrix, enhancing model stability. A selection of variables that correlate with the response is then extracted, and these are utilized to construct a linear model. Ultimately, this linear model is seamlessly integrated into the nonlinear regression framework. Since SR maintains the good performance of the nonlinear model and the interpretability of the linear model with a low level of computational complexity, it effectively improves the accuracy and robustness of the LIBS quantitative analysis medium and also ensures the rapidity of modeling and prediction in the LIBS system [106].

Song et al. experimentally demonstrated that synergistic regression (SR) outperformed PLSR and KELM in all quantitative tasks, and it showed significant improvement in the tasks of sulfur, volatile matter, and ash. The coefficients of determination R^2 for calorific, volatile, and ash analyses were 0.960, 0.944, and 0.923, respectively, and the MAEs for calorific, volatile, and ash analyses were 0.299, 0.590%, and 0.855%, respectively [106].

(5) Deep learning

Deep learning, an advanced form of machine learning, harnesses the synergies between enhanced computational capabilities and specialized neural network architectures to discern intricate patterns within vast datasets. In scenarios involving extensive data, conventional decision-making frameworks often stumble upon inconsistencies or even the absence of definitive rules, posing significant challenges for traditional information processing techniques. Conversely, neural networks excel at navigating such complexities, offering insightful recognition and assessment capabilities that are commensurate with the task's demands [107]. In addition, big data can make traditional chemometrics quite inefficient and even overwhelming, but it is completely favorable to neural network methods. Therefore, neural networks are suitable and powerful tools for LIBS data processing and analysis [108].

Artificial neural network (ANN) is a widely used machine learning model originating from biology, combining mathematical and physical methods to abstract the neural network of the human brain from the point of view of information processing to establish a self-adaptive nonlinear dynamic system with a strong input-output nonlinear mapping ability, self-adaptive ability, and learning ability. ANN, a parallel distributed system, revolutionizes the landscape of artificial intelligence and information processing by embracing a fundamentally distinct mechanism. It transcends the limitations of traditional logic-centric AI in managing intuitive and unstructured data, effectively addressing their shortcomings. ANN boasts self-adaptability, self-organization capabilities, and real-time learning prowess, thereby offering unparalleled advantages in handling complex information scenarios [109]. Feedforward and multilayer perceptron ANN consists of a set of interconnected neurons that are organized into layers. The ANN architecture comprises an input layer, which introduces external data without computation, a series of hidden layers sandwiched between, and an output layer. The hidden layers encapsulate neurons, each equipped with n inputs, each weighted, and a bias factor among the weighted inputs. The activation function modulates the influence of these weighted inputs and biases on the neuron's output, sometimes acting as thresholds. The learning mechanism revolves around fine-tuning the inter-node weights through various algorithms. Fundamentally, an ANN operates as a functional transformation, translating an input vector into a corresponding output vector. Obviously, in order to make accurate predictions, ANNs need to have an appropriate number of neurons in each hidden layer with appropriate activation functions and a large number of learned data points [110]. Currently, the research work on ANN applied to LIBS data analysis focuses on two aspects: (1) selecting different spectral data as input variables for the ANN model, including the full spectrum, specific spectral bands, and manually or algorithmically extracted characteristic lines and more; and (2) optimizing the ANN configuration parameters, including the conventional combinatorial optimization, the use of genetic algorithms (GA) and particle swarm algorithms (PSO) for global optimization search and so on. Under the optimal input and optimal configuration, the ANN maintains

an efficient learning state when processing LIBS data and obtains optimal prediction results, which is conducive to the application of LIBS technology in practical scenarios in multiple domains, and it can effectively remove the noise and redundant information, and improve the signal-to-noise ratio and reliability of the data.

Zhang et al. used the ANN model for analyzing 40 standard coal samples, with a strategic division of 30 samples allocated for calibration purposes, while the balance of 10 samples served as validation instances for predictive accuracy assessment. For the predicted samples, the MAEs of ash, volatile matter, and calorific value were determined to be 0.69, 0.87, and 0.56, the RMSEPs were 1.086, 1.128, and 0.694, and the ARSDs were 2.15, 1.39, and 0.22 [90].

Yao et al. used LIBS for rapid quality analysis of pulverized coal, combined with clustering analysis and an ANN approach to devise a predictive model. They then harnessed a genetic algorithm (GA) optimization strategy to fine-tune the ANN's parameters, ensuring optimal weights and thresholds that accurately reflect the correlation between coal quality and its plasma spectral characteristics. The results showed that the MAEs of the validation set for ash, volatile matter, fixed carbon, and total calorific value were 0.82%, 0.85%, 0.96%, and 0.48 MJ/kg, respectively. The mean standard deviations (MSDs) of the duplicate samples were 1.64%, 0.92%, 1.08%, and 0.86 MJ/kg, respectively, which demonstrated high inter-sample reproducibility. This rapid coal analyzer is capable of analyzing coal quality reliably and accurately [111].

Lu et al. used a combination of ANN and GA for the rapid determination of the total calorific value (GCV) of coal. An ANN model was constructed for quantitative LIBS analysis, aiming to diminish matrix effects and nonlinearity. This was complemented by a genetic algorithm integration, which somewhat mitigated ANN's random influence, ensuring a concise yet effective approach. The results showed that the MSD of GCV for 50 samples of the prediction set was 0.38 MJ/kg in four trials, demonstrating that the ANN model can provide high modeling reproducibility in GCV analysis. The MAE of GCV for the prediction set was 0.39 MJ/kg [112].

(6) Transfer learning

LIBS integrated with machine learning methodologies exhibits robust potential for precise quantitative elemental analysis. However, when disparities arise in the distribution of training and test datasets owing to variations in measurement conditions and sample compositions, compounded by the challenges of limited sample availability due to the high cost and extended duration of analyte content certification, the accuracy and reliability of the machine learning models are compromised [97]. Transfer learning is an approach that enhances learners in a specific domain (target domain) by leveraging knowledge from another, albeit related, domain (source domain), thereby tackling the issue of limited training data availability [113]. Transfer learning has been implemented in various LIBS contexts to address challenges like matrix effects and scarcity of certified samples for authentication purposes, enhancing the technique's versatility and effectiveness [114].

Chen et al. pre-trained a neural network on both source and target domain training sets utilizing a hybrid transfer learning approach (HTr-LIBS), which integrates fine-tuning with sample reweighting. This method iteratively adjusted the sample weights in the source domain based on prediction errors, subsequently refining the pre-trained neural network with the optimally weighted samples derived from the target domain training set. The experiment introduced 310 source-domain samples and used 20 target-domain samples for quantitative analysis. When the size of the training set reached 19, the R^2 of the ash content and volatile content of HTr-LIBS were 0.9029 and 0.9627, and the RMSEP was 2.0588% and 1.6083, which significantly improved the analytical accuracy of the proposed method [114].

Cui et al. used a multi-task regularized convolutional neural network (TrMR-CNN) to analyze the mordant composition. TrMR-CNN is a convolutional layer that employs a pre-existing neural network model as its foundation and undergoes training using a limited quantity of target domain training samples. Multi-task regularization is incorporated to

harness prior knowledge of sample composition, thereby imposing constraints on the source model to enhance its relevance and performance. This allows for better analysis of coal datasets with limited sample sizes and varying ranges of analyte concentrations. Based on the quantification results of the 10-sample target training set, the predicted MAEs for ash, volatile matter, and sulfur are 3.335, 1.371, and 0.408, and the RMSEPs are 0.278, 0.169, and 0.42. Compared with PLSR, SVR, and non-transported CNN models, the proposed method RMSEP decreased by 19.9%, 5.9%, and 7.7%, respectively. The results show that the effectiveness and robustness of the proposed method are demonstrated in ash, volatility, and sulfur analysis. The accuracy and robustness of limited datasets outperform the baseline method and are capable of handling a variety of related quantitative tasks [97].

(7) Algorithmic combination

According to the different principles of various algorithms and the different conditions under which the algorithms are applicable, various algorithms are applicable to different degrees in the application of LIBS, and by combining the advantages of different algorithms, better performance than a single algorithm can be obtained. This includes improving performance, dealing with complex problems, enhancing robustness, resolving limitations, accelerating computation, and improving interpretability. However, how to effectively combine different algorithms and fully utilize their advantages is a challenging problem that needs to be selected and adapted according to the characteristics of the specific problem and dataset, which largely depends on the characteristics of the specific algorithms and how they can be used in combination. Some examples of algorithmic combinations in LIBS detection media are listed below.

High-precision quantitative analysis of material composition based on laser-induced breakdown spectroscopy requires large sample sizes for accurate modeling. However, in most cases, abundant standard samples are not available. Li et al.'s small sample machine learning (SSML) algorithm based on PCA-PLS partitions sample spectra into enriched subsets through partial data extraction to enhance quantitative analysis of coal using LIBS. In the new prediction model with a training sample size of 15, the average relative error (ARE) of carbon prediction is less than 4%, and the R^2 and RMSEP are 0.77 and 1.7869 wt% for carbon, 0.70 and 2.1502 wt% for ash, 0.73 and 0.7741 MJ/kg for calorific value when the number of training samples is 6 and the number of predicted samples is 94. The R^2 and RMSEP are 0.77 and 1.7869 wt%, respectively. The ARE and RMSEP of the new model were reduced by about 50% compared to the conventional prediction model, which can be attributed to the following reasons: (1) The new algorithm used higher quality spectra in the modeling, which provided stable and representative data for model training. (2) A strengthened correlation was observed between the pertinent emission spectra and predicted outcomes, achieved by augmenting the training dataset through multiple extractions and permutations of spectra from individual samples. This approach facilitated the extraction of more profound information from the available spectral data, even with a reduced sample size [35].

Dou et al. selected three partitioning methods, random selection (RS), Kennard–Stone (KS), and sample partitioning based on joint X-Y distance (SPXY), in combination with quantitative models (PLSR, SVR, and RF), to analyze and compare the model performance and prediction accuracy of the methods. The results showed that the SPXY dataset partitioning combined with the RF model had better prediction performance than PLSR and SVR models combined with RS and KS methods. The ash content R^2 of RF and SPXY methods was 0.9843, RMSEP was 1.3303, and MRE was 7.47%. The volatile fraction R^2 was 0.9801, RMSEP was 0.7843, and MRE was 2.19%. The calorific value R^2 was 0.9844, the RMSEP was 0.7324, and the MRE was 2.27% [115].

4.3. LIBS in Conjunction with Other Technologies

LIBS, an elemental analysis method rooted in atomic emission spectroscopy, harnesses the power of lasers as an excitation source. Widely adopted for elemental analysis and coal quality assessment, LIBS boasts simplicity in operation, minimal sample preparation

requirements, simultaneous multi-element analysis capabilities, and high sensitivity. Nevertheless, the intricacies of coal's composition and molecular architecture pose challenges. Coal quality analysis transcends mere elemental analysis; it encompasses molecular structure and composition as well. While LIBS efficiently captures atomic spectral signatures from coal, its inability to comprehensively capture molecular spectral information may result in analysis inaccuracies stemming from incomplete data [116]. To elevate its analytical prowess, LIBS technology is fused with complementary methods like LIBS-Raman, enabling a more exhaustive understanding of the material's composition by accessing a broader spectrum of information [117], and laser ablation-LIBS used to reduce matrix effects [118]. This enhancement bolsters the precision and dependability of the analysis outcomes [104].

(1) XRF-assisted LIBS based on high stability

X-ray fluorescence spectrometry (XRF) shows good reproducibility in coal analysis [119]. XRF operates on the principle that individual atomic constituents, when stimulated by external energy, emit secondary X-ray fluorescence with a distinctive energy signature (ED-XRF) or wavelength (WD-XRF). This fluorescence intensity serves as a basis for quantifying the element. However, ED-XRF's applicability is limited to inorganic ash-forming elements with atomic numbers exceeding 11, excluding organic constituents like C and H. Consequently, it falls short in determining coal's calorific value and volatility [120]. LIBS has relatively low measurement repeatability, but it is capable of analyzing all the key elements in coal. In contrast, while XRF's focus lies solely on ash-forming elements, it excels in providing analysis with remarkable stability and reliability [121]; in stark contrast, XRF, despite its limitation to ash-forming elements, boasts exceptional stability. By integrating LIBS and XRF, we created a hybrid approach that retains LIBS' unique capability to directly analyze organic elements like C and H in coal, while leveraging XRF's strength in ensuring stability for the determination of other inorganic ash elements, thereby addressing LIBS' inherent instability in this regard [122].

The study by Tian et al. is based on the highly stable XRF-assisted LIBS analysis method, principal component analysis to screen out the principal components, and PLS to establish the coal quality prediction model using LIBS-XRF coal quality analyzer, which consists of a LIBS module, an XRF module, a sample conveying module, a control module, and an operating software. The experimental outcomes demonstrated that the RMSEP for the coal's calorific value, ash content, volatile matter, and sulfur content amounted to 0.62 MJ/kg, 1.46%, 0.23%, and 0.19%, respectively, while the mean standard deviation (SD) remained unchanged at 0.11 MJ/kg, 0.49%, 0.15%, and 0.09%, respectively [120].

In addition, Tian et al.'s study based on the high stability of the XRF-assisted LIBS analysis method, the PCA method to screen out the principal components, and the use of PLS to establish the coal quality prediction model of the experimental results show that the calorific value of the method, the ash, the volatile content and the main elements of the C of the R^2 are all more than 0.95, in particular, the calorific value of the calorific value, the ash, the volatile content of the SD are 0.11 MJ/kg, 0.17%, and 0.41%, and the RMSE of elemental analysis is less than 2%, and the RSD is less than 4% [122].

(2) Synergy of Fourier transform infrared spectroscopy and LIBS information

In Fourier transform infrared spectroscopy (FTIR), the light source is a continuous wavelength light source, the sample generates an interferogram after infrared absorption, and the interferogram is converted into a spectrogram by Fourier transform. The merits of FTIR encompass swift scanning capabilities coupled with high resolution, substantial luminous flux yielding enhanced sensitivity, an extensive spectral range, and unparalleled measurement precision. The principle is that the molecules in the sample are illuminated at continuous wavelengths; their dipole moments change and absorb infrared spectra at specific wavelengths, which can be obtained after detection by a detector and analog-to-digital conversion.

FTIR boasts rapid scanning, high resolution, intense luminous flux, and remarkable sensitivity, spanning a wide spectral range with accurate measurements. For synergy with LIBS, their spectral matrices are vertically stacked to form an integrated fusion matrix. Usually, preprocessing of the spectra is required before fusion. First, the LIBS and FTIR spectral data are preprocessed separately using a normalization method, and subsequently, the preprocessed spectral data are merged into a single matrix [104].

He et al. performed coal quality analysis based on the mutual information-particle swarm optimization-kernel extreme learning machine model (MI-PSO-KELM) for FTIR and LIBS fusion. The procedure is to first obtain the fused spectral matrix, and the preprocessed LIBS and FTIR spectral matrices are directly connected end-to-end. Subsequently, MI is employed to prune redundant variables from the spectral data, and PSO is leveraged to refine the retained variables, resulting in a subset that exhibits enhanced prediction accuracy. Finally, coal quality analysis was performed based on KELM. During the modeling phase, a subset of 30 coal samples was randomly chosen from a total of 45 samples to serve as the calibration set, while the remaining 15 samples comprised the test set. The results showed that the prediction result of ash R^2 was 0.9821, and RMSEP was 0.9687%. The R^2 of volatile matter prediction was 0.9789, and RMSEP was 1.3218% [104].

Yan et al. used the marine predator algorithm to optimize the kernel extreme learning machine model to combine the synergistic effect of FTIR and LIBS information for coal quality analysis. The results show that the sum R^2 of carbon, ash, volatile matter, and calorific value are higher than 94.1, and the RMSEP values are lower than 1.7% or 1.7 MJ/kg, compared with the analysis of LIBS or FTIR by MPA-KELM model alone. The method has great potential for predicting coal carbon, ash, volatile matter, and calorific value. The method has great potential in predicting coal carbon, ash, volatile matter, and calorific value [123].

(3) Synergy of near-infrared reflectance spectroscopy and LIBS information

Near-infrared reflectance spectroscopy (NIRS) is a rapid technique for analyzing molecular structures because NIRS collects information mainly from overtones and combined bands generated by molecular vibrations and offers a range of advantages such as time savings, safety, and no need for sample preparation [124].

Based on the aforementioned analysis, Yao et al. devised a quantitative assessment model for calorific value, volatile matter, and ash content by integrating LIBS and NIRS data, utilizing the PLS method for model construction. The results showed that the RMSEP of calorific value, volatile fraction, and ash were 0.192 MJ/kg, 0.672%, and 0.849%. The calorific value and volatile fraction results are superior to the LIBS-based model because they have almost equal correlations with elements and molecules; the ash analysis results are inferior to the LIBS-based model because most of the minerals have a small response to the near-infrared (NIR) spectra [124].

(4) Single-beam splitting technology

Cheng et al. used single-beam split LIBS (SBS-LIBS) for quantitative analysis of coal, where the laser pulse was divided into two parallel beams that converged onto a single spot on the sample's surface, resulting in a notable amplification of the spectral intensity compared to conventional LIBS techniques. A quantitative analysis model was formulated utilizing the support vector machine approach. RMSEP, average prediction correlation error (AREP), and average absolute error of prediction (AAEP) were used to evaluate the accuracy of SBS-LIBS on SVM. The findings indicated a comprehensive enhancement in the analysis of coal utilizing SBS-LIBS, as evidenced by the reduction in AAEP for carbon, hydrogen, nitrogen, ash, and volatile fractions from 1.04%, 0.18%, 0.08%, 1.80%, and 4.30% to 3.70%, 0.27%, 0.09%, 3.06%, and 7.99%, respectively [125].

4.4. Analysis and Summary

(1) Principal component regression

In applications, PCR can effectively deal with the problem of multicollinearity among independent variables. In multiple regression analysis, when there exists a high degree of collinearity between independent variables, it may lead to unstable estimation of regression coefficients. PCR avoids the effect of multiple covariance by converting the original independent variables into uncorrelated principal components through PCA and selecting a part of them as the new independent variables for regression analysis. Moreover, PCR can simplify the complexity of the regression model by transforming high-dimensional data into low-dimensional data through dimensionality reduction operations. This helps to reduce the amount of computation, improve the interpretability of the model, and reduce the risk of overfitting.

However, since PCR uses principal components as independent variables for regression analysis, and these principal components are usually linear combinations of the original independent variables, it is difficult to directly interpret the meaning of each principal component. This may lead to limited interpretability of the model, especially in cases where an in-depth understanding of the effect of the independent variables on the dependent variable is required. Moreover, PCR may lose some of the original information in the dimensionality reduction process. While this helps simplify the model and reduce the risk of overfitting, it may also lead to a decrease in the predictive power of the model in some aspects. Finally, the performance of PCR depends greatly on the number and type of principal components selected. If too many or too few principal components are selected, this may affect the predictive ability of the model. In addition, for different datasets and problems, selecting the optimal number of principal components may require several attempts and comparisons through methods such as cross-validation. And as a large number of different types of coal are used in boiler power plants, PCR may not be able to fulfill this need well.

Currently, PCR modeling is mostly used for media classification to improve boiler combustion efficiency and for spectral data processing to withdraw from researchers' studies.

(2) Partial least squares

PLS has emerged as a highly adaptable technique for multivariate data analysis, characterized as a supervised approach tailored to excel in predicting outcomes in multivariate contexts. Its core lies in identifying pertinent linear subspaces within the explanatory variables rather than focusing solely on the variables themselves, thereby enhancing predictive accuracy [126]. In contrast to the PCR model described above, PLS calculates the principal components of the raw spectral data in a way that, in addition to considering the maximum variance of the principal components of the spectral data, also takes into account the maximum degree of correlation between the principal components and the concentration of the analyzed components, linearly decomposes the attributes to be measured as well, extracts the effective factors, and then combines them with the principal components in the spectral information in order to find the optimal prediction model. In this way, the correlation between spectral data and component concentrations can be maximized while retaining sufficient information about the original spectral data. And the PLS model is relatively simple to construct and small in computation, which can process the spectral data quickly.

However, PLS, being a projection-based method, inherently disregards dimensions in the variable space that are dominated by uncorrelated noise variables. Consequently, for predictive purposes, variable selection may not appear essential since PLS inherently adjusts variable weights, both upwards and downwards. Nevertheless, an exceedingly high number of independent variables coupled with a limited sample size can still compromise the accuracy of PLS regression outcomes [126]. From a prediction point of view, a large number of uncorrelated variables may make a big difference to the test set prediction [127].

The rationale behind this is that as the variable count escalates, the PLS algorithm confronts growing challenges in determining the appropriate dimension of the correlation subspace within the p-dimensional variable space commensurate with the number of independent variables. While PLS variable selection can bolster model performance, it may concurrently strip away valuable redundancies within the model. Moreover, relying on a limited set of variables for prediction underscores the significant impact each of these variables exerts on the final model [128]; the mechanism of jumping in LIBS spectra is that elements present in the substance can be excited to form peaks, so intermittent undulations of very low intensity will often be considered noise, which contains a great deal of useless information that can have an effect on the PLS-fitted spectra.

(3) Machine learning

Machine learning possesses a robust capacity to tackle intricate problems through example-based learning, iteratively examining data to discern intricate correlations between target and input variables [129]. However, without a thorough comprehension of both the problem at hand and the intricacies of the machine learning algorithm employed, it frequently results in a reliance on a black-box approach, potentially leading to biased outcomes [110].

As an outstanding representative of machine learning algorithms, neural networks are well suited to solve complex, ill-defined, highly nonlinear problems with many different and random variables [108]. The correlation between LIBS spectra and analyte quantities exhibits a degree of nonlinearity, stemming from the saturation phenomenon in the signal (arising from self-absorption of intense plasma emission lines) and the chemical matrix influence inherent in coal samples [130]. The application of deep learning relative to traditional methods in LIBS, including preprocessing, feature extraction, data justification, model transfer, multiple outputs, classification, and regression, and the deep learning methods show superiority over the traditional methods in all of these aspects, demonstrating the potential and validity of the application of deep learning in LIBS analysis [131]. The implementation of deep learning in LIBS analysis, encompassing preprocessing, feature extraction, data refinement, model adaptability, multi-output generation, classification, and regression, outperforms traditional methodologies in all these facets, underscoring the potential and validity of deep learning's application in enhancing LIBS analysis [132].

Currently, it is prevalent for machine learning modeling procedures to acquire spectral data repetitively and subsequently calculate a mean spectrum per sample, aiming to mitigate variations stemming from experimental parameter fluctuations and sample inhomogeneity [35]. As a consequence, only a singular spectrum per sample emerges, implying that each spectrum within a sample's spectral dataset is utilized solely once. Traditional model training methodologies typically partition samples into training and testing sets randomly, primarily centered on the averaged spectrum. When the sample size dwindles to the lower teens or even single digits, the model starts to deviate due to overfitting during training, stemming from an inadequate amount of validation data. In essence, the model struggles to grasp the spectral attributes of the samples and the underlying patterns between the metrics [133], and the predictive performance of machine learning as well as linear analysis methods for them drops dramatically.

KELM is an improved ELM algorithm proposed by introducing a kernel function on the basis of ELM. KELM not only inherits the advantages of ELM, such as good generalization performance and fast learning speed but also due to the introduction of the kernel function, so that KELM can obtain the least-square optimization solution so that KELM has a more stable and superior performance. Compared with the support vector machine and the basic ELM algorithm, it has better stability and stronger generalization performance, so the kernel extreme value learning machine has been widely used in classification and regression problems. However, the classification performance of the KELM algorithm relies heavily on the choice of kernel function and regularization parameters. If they are not chosen properly, it may lead to poor classification performance of the algorithm. Therefore,

a suitable choice of kernel function and regularization parameters is needed to obtain better performance.

As for ANN in machine learning, the model demonstrates superb data mining ability in LIBS spectral analysis, which can effectively overcome and utilize noise, matrix effect, baseline interference, self-absorption, and laser energy fluctuation, and at the same time, it can extract valuable nonlinear information from all kinds of raw data for adaptive learning to improve the accuracy and sensitivity of LIBS analysis. Recently, the conjunction of ANN and other algorithms has also been the focus of research, complementing the information of ANN with various types of techniques to improve detection accuracy. However, the application of ANN in LIBS data analysis is developed from a shallow simple network to deep learning, and for high-dimensional and data-volume spectral data, the shallow neural network may not be able to learn the spectral characteristics deeply enough to meet the accuracy requirements.

Domain adaptation research is driven by the observation that humans possess the innate ability to leverage prior knowledge to tackle novel challenges more efficiently or devise superior solutions. Migration learning facilitates the calibration and prediction of data across diverse domains, tasks, and distributions, enabling the extraction of valuable insights from the source problem to effectively address a distinct yet interconnected target problem [134]. Relative to other models, migration learning is particularly suitable for small samples or data scarcity, and its ability to leverage existing knowledge and resources reduces the dependence on the amount of data and improves the model's generalization ability and robustness.

However, in migration learning for quantitative analysis, the disparity in distributions between the source and target domains results in a mixed bag of data from the source domain. Some of these data may prove beneficial for learning in the target domain, while others may be irrelevant or even detrimental. To mitigate the negative impact of harmful source data and amplify the contributions of useful ones, the process iteratively adjusts the weighting of source domain data [134]. When the correlation between the source and target tasks is not strong, transfer learning can lead to negative migration, where the migrated knowledge negatively affects the new task. Migration learning models are often more complex and, therefore, require careful tuning of the relationship between the source and target tasks to ensure effective knowledge migration, but this can make debugging and optimization of the model more difficult.

(4) Modeling time

When using different models for media composition analysis of LIBS data, modeling time may not be the primary metric to assess the performance of the model, and more attention is paid to the accuracy of the model. In practical industrial applications, hundreds of coal samples are required to construct calibration models with high adaptability and robustness. However, the modeling time increases with the amount of spectral data, which poses a great challenge to the computational resources [90]. Modeling time is certainly an area of concern during the experimental testing phase. A model with a shorter modeling time can significantly improve the experimental efficiency. However, researchers seldom explicitly mention the modeling time of the models they employ. This may be because modeling time is not the main criterion for traditionally evaluating the performance of a model, and the impact of modeling time on the overall study is not significant. However, for application scenarios pursuing efficient experimental processes and fast responses, modeling time is certainly an aspect that deserves further exploration and optimization.

In Zhang et al.'s study, 40 standard coal samples were measured, and their analysis compared the modeling time of four algorithms, PCA, PLSR, SVR, and ANN, in terms of ash, volatile matter, and calorific value. It was found that PLSR and PCA had a higher modeling efficiency, with modeling times of 8 s and 10 s for all three components, and the modeling times of ANN for the three components were 94 s, 71 s, and 86 s, respectively, while SVR has the longest modeling time, with times of 1120 s, 1818 s, and 2099 s, respectively. The reason for this is that the two algorithms, PLSR and PCA, have a relatively simple

training process, require fewer parameters to be optimized and fewer computations, and are unable to construct a nonlinear relationship between the strength of the LIBS eigenline and the analytical parameters. The ANN is characterized by a vast quantity of weights, necessitating frequent updates through the error backpropagation algorithm. This iterative process significantly slows down the training speed of the network. Conversely, SVR modeling experiences the longest duration due to the intensive search for the optimal values of its two crucial parameters, a time-consuming endeavor in the nonlinear modeling framework [90]. Researchers can roughly deduce the length of the modeling time based on the principle of the algorithm so as to select the appropriate algorithm to improve the efficiency of the experiment, and for practical industrial applications, selecting the appropriate algorithm not only improves the accuracy of the measurement but also reduces the load on the computer and shortens the detection time.

5. Conclusions and Prospects

At present, countries are vigorously developing renewable energy to alleviate the global carbon crisis. However, coal still has an indispensable position in the fields of iron and steel, electricity, and chemical industry. Under the study of adsorption of carbon dioxide, nitrogen oxides, and other emission gases produced by coal combustion, the environmental damage caused by coal combustion has been reduced. In the context of coal combustion, the realization of rapid detection of coal or real-time detection of ash content, volatile matter, and calorific value, and optimization of coal allocation can improve the combustion rate of coal-fired power plants at the same time but also reduce the emission of pollutants. Therefore, the development of rapid detection equipment for media has become the focus of research in recent years. LIBS has the ability of high sensitivity and rapid analysis for multi-element detection, which is suitable for elemental analysis of complex media composition. However, it may lead to multiple sources of errors at the experimental and principle levels, which reduces the accuracy of the measurement. For this paper, the principle of LIBS and the pressure-cake experimental process are described in detail, the error analysis at the experimental and principle levels is carried out, the algorithmic model for fitting spectral data and elemental compositions is summarized, the detection is applied in conjunction with other techniques, and the advantages and disadvantages of the various algorithms are analyzed.

At present, the main research direction of researchers on LIBS in media detection is pressure cake detection, which will increase the complexity of the equipment, and the sample preparation is difficult, increasing the time of media detection. In order to realize real-time monitoring of coal and provide timely and effective guidance for coal combustion, the development of online LIBS technology is particularly important. At present, online LIBS for media analysis is mainly for coal particle flow detection, but its laser action mechanism is different from that of the pie-pressing type. After pressing the coal cake into a cake, it can be considered that the laser acts on the solid, but the coal particle flow is a state of gas–solid two-phase flow, which increases the instability of the sample. Therefore, the detection of coal particle flow by LIBS is not only the focus of the next research but also the focus of the application of LIBS detection medium.

In addition, combining LIBS technology with other on-line analytical techniques (NIRS, microwave transmission method, etc.) is also a solution to improve the prediction accuracy of coal quality indicators. To effectively apply LIBS technology in practical settings, it is imperative to integrate the LIBS measurement system with combustion optimization and control mechanisms. This integration aims to optimize coal consumption, mitigate pollutant emissions, and bolster the safety of industrial processes. In essence, coal quality analysis represents a promising arena and market for LIBS technology, offering substantial benefits and growth opportunities [31].

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